2023 力学：

1. $\vec{x} = 4t^2\vec{i} + (2t+3)\vec{j}$

$\vec{v} = 8t\vec{i} + 2\vec{j}$ $\quad\therefore \vec{v}|_{t=1} = 8\vec{i} + 2\vec{j}$

2. 径向：$a_r = \ddot{r} - r\dot{\theta}^2 = \frac{d^2r}{dt^2} - r\cdot(\frac{d\theta}{dt})^2$

切向：$a_\theta = 2\dot{r}\dot{\theta} + r\ddot{\theta} = 2\cdot\frac{dr}{dt}\cdot\frac{d\theta}{dt} + r\frac{d^2\theta}{dt^2}$

3. $\Delta p = \int_0^1 F\cdot dt = c\cdot\int_0^1 t^2\cdot dt = \frac{c}{3}$

$\therefore \Delta\vec{p} = \frac{c}{3}\vec{i}$.

4. $kV_m^2 = mg \quad\therefore V_m = \sqrt{\frac{mg}{k}}$

5. 设面密度为 ρ 对于半径为 R 的半圆盘：$2\pi r_C'\cdot\frac{1}{2}\pi R^2 = \frac{4}{3}\pi R^3$.

$\therefore r_C' = \frac{4R}{3\pi}$

$r_C'' = \frac{4r}{3\pi}$.

$\therefore r_C = \frac{\frac{\pi R^2}{2}\cdot\rho\cdot\frac{4R}{3\pi} - \frac{\pi r^2}{2}\cdot\rho\cdot\frac{4r}{3\pi}}{\rho(\frac{\pi R^2}{2} - \frac{\pi r^2}{2})} = \frac{4}{3\pi}\cdot\frac{R^2+Rr+r^2}{R+r}$.

6. $J = \frac{1}{2}mR^2$.

动能定理：$mgh = \frac{1}{2}mV_C^2 + \frac{1}{2}\cdot(\frac{1}{2}mR^2)(\frac{V_C}{R})^2$.

$= \frac{1}{2}mV_C^2 + \frac{1}{4}mV_C^2 = \frac{3m}{4}V_C^2$.

$\therefore V_C = \sqrt{\frac{4gh}{3}}$ #.

7. $\vec{F} = 5t^2\vec{i} - 6\vec{j}$.

$\vec{v} = 10t\vec{i} + 0\vec{j} \quad\therefore \vec{p} = m\vec{v} = 60t\vec{i} + 0\vec{j}$

$t=1$ 时：$\vec{F} = 5\vec{i} - 6\vec{j}$

$\vec{p} = 60\vec{i} + 0\vec{j}$

$\therefore \vec{M} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 5 & -6 & 0 \\ 60 & 0 & 0 \end{vmatrix} = 360\vec{k}$

$\therefore$ 大小为 $360\ kg\cdot m^2\cdot s^{-1}$ #.

8. 设单位时间喷出质量为 w.

初: 速: V 质量: M.

末: 火箭: 速: $V+\Delta V$ 质量: $M-w\Delta t$.

燃料: 速: $V+\Delta V+u$ 质量: $w\cdot\Delta t$.

动量守恒: $MV = (V+\Delta V)(M-w\Delta t) + (V+\Delta V+u)w\Delta t$. 略去=所小量.

$$\Delta V\cdot M = -uw\Delta t = -u\cdot\Delta M$$

取极限: $dv = -u\cdot\frac{1}{M}dM$.

$$\therefore \quad v-v_0 = -u\ln M\Big|_{M_0}^{M_S}, \quad v_0=0.$$

$$\therefore \quad V = u\cdot\ln\frac{M_0}{M_S}. \qquad \therefore \quad M_S = e^{-\frac{V}{u}}\cdot M_0.$$

进而 $m = M_0 - M_S = (1-e^{\frac{-V}{u}})M_0$ #.

9. 多普勒效应: 波源运动:

开来: $440 = \frac{V_{声}}{V_{声}-V}f$. ; $V_{声}=330$.

开走: $392 = \frac{V_{声}}{V_{声}+V}\cdot f$

解出 $V = \frac{495}{26}$ m/s.

10. 令 $x=1$ $\therefore$ $y = A\cos(wt+k)$.

11. $x^2+y^2 = A^2(\sin^2wt+\cos^2wt) = A^2$

一个半径为 A, 圆心为原点的圆.

运动方向逆时针.

12. 阻尼: $\beta>w$, 则: $x = C_1e^{(-\beta-\sqrt{\beta^2-w^2})t} + C_2e^{(-\beta+\sqrt{\beta^2-w^2})t}$.

临阻尼: $\beta<w$, 则: $x = e^{-\beta t}\left[C_1\cos\sqrt{w^2-\beta^2}\,t + C_2\sin\sqrt{w^2-\beta^2}\,t\right]$

大题：

T1：解：$v^2 = 2gx$，$x = \frac{L}{2}$ $\therefore v = \sqrt{gL}$.

$\therefore V_C = \frac{\frac{m}{2}\cdot\sqrt{gL}}{m} = \frac{1}{2}\sqrt{gL}$，方向竖直向下.

$N_1 = \frac{1}{2}mg$. 设线密度为 ρ，$\rho = \frac{m}{L}$.

取一小段质量元 dm，$dm = \rho dx = \frac{m}{L}\cdot dx$.

动量定理：$N_2\cdot dt = dm\cdot v = v\cdot\frac{m}{L}\cdot dx$.

$\therefore N_2 = v\cdot\frac{m}{L}\cdot\frac{dx}{dt} = v^2\frac{m}{L} = mg$.

$\therefore N = N_1 + N_2 = \frac{3mg}{2}$ #.

方向竖直向上.

T2：1）由 $\frac{dE_p}{dr} = -f = -\frac{k}{r^2}$.

$\therefore dE_p = -\frac{k}{r^2}\cdot dr = k\cdot d\frac{1}{r}$

$\therefore E_p = k\cdot\int_{\infty}^{r} d(\frac{1}{r}) = \frac{k}{r}$ #.

2）：设最近距离为 d，最终速度为 v，该过程角动量守恒.

$\therefore v_0\cdot b = dv$ ①

远处势能为0，近处势能为 $\frac{k}{d}$.

$\therefore \frac{k}{d} = \frac{m}{2}(v_0^2 - v^2)$ ②.

①、② 解出：$\begin{cases} d = \dfrac{mb^2v_0^2}{k+\sqrt{k^2+v_0^4b^2m^2}} \\ v = \dfrac{-k+\sqrt{k^2+v_0^4b^2m^2}}{mbv_0} \end{cases}$

T3: 解: 考虑其中一个杆: $\Delta h = \frac{1}{4}l$.

$$\therefore \begin{cases} \frac{1}{4}mgl = \frac{1}{2}mV_C^2 + \frac{1}{2}J\omega^2 \quad ① \\ J = \frac{1}{12}ml^2 \quad ② \\ V_C = \frac{l}{2}\cdot\omega \quad ③ \end{cases}$$

解出 $\omega = \sqrt{\frac{3g}{2l}}$. $\therefore V_{杆} = \omega\cdot l = l\cdot\sqrt{\frac{3g}{2l}} = \sqrt{\frac{3gl}{2}}$

计算质心加速度: 采用极坐标方法: $a_C = r'\cdot\ddot{\theta} = \frac{l}{2}\cdot\beta$.

而 $I\beta = mg\cdot\frac{l}{2}$, $I = \frac{1}{3}ml^2$ $\therefore a_C = \frac{3g}{4}$ $\therefore mg - N_1 = ma_C$

$\therefore N_1 = \frac{mg}{4}$ $\therefore N = 2N_1 = \frac{mg}{2}$.

T4: 解: 动量守恒: $mV_0 = -mV_1 + 2mV_2$. ① 向右为正方向.

角动量守恒: $mV_0\cdot\frac{\sqrt{2}l}{4} = -mV_1\cdot\frac{\sqrt{2}l}{4} + J\omega$. ②

$$J = m\cdot(\frac{l}{2})^2 + m\cdot(\frac{l}{2})^2 = \frac{ml^2}{2}. \quad ③$$

能量守恒: $\frac{1}{2}mV_0^2 = \frac{1}{2}mV_1^2 + mV_2^2 + \frac{1}{2}J\omega^2$. ④.

$$\sqrt{2}V_2 = \omega l. \quad ⑤$$

$$解出\ \omega = \frac{4\sqrt{2}V_0}{7l}.$$

T5: 1): 设 原点处 $u(0,t) = A\cos(\omega t + \varphi)$. $\therefore \varphi = 0$.

$$= A\cos(\omega t) = A\cos\left(\frac{2\pi}{T}\cdot t\right)$$

$$\therefore u(x,t) = A\cos\left(\frac{2\pi}{T}(t - \frac{x}{V})\right) \quad 而 V = \frac{\lambda}{T}.$$

$$= A\cos\left(\frac{2\pi}{T}\cdot t - \frac{2\pi x}{\lambda}\right).$$

2): $x = \frac{\lambda}{2}$ $\therefore u(\frac{\lambda}{2},t) = A\cos(\frac{2\pi}{T}\cdot t - \pi)$.

3): $x = \frac{\lambda}{4}$ $\therefore u(\frac{\lambda}{4},t) = A\cos(\frac{2\pi}{T}t - \frac{\pi}{2})$.

$$V(\frac{\lambda}{4},t) = -\frac{2\pi}{T}A\cdot\sin(\frac{2\pi}{T}\cdot t - \frac{\pi}{2}), \quad t = \frac{T}{2}$$

$$V = -\frac{2\pi}{T}A, \text{ 方向向下}.$$